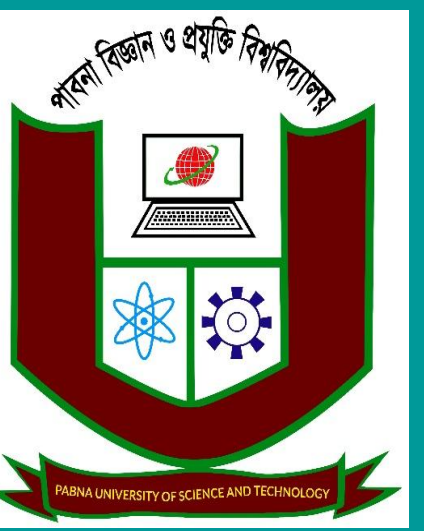


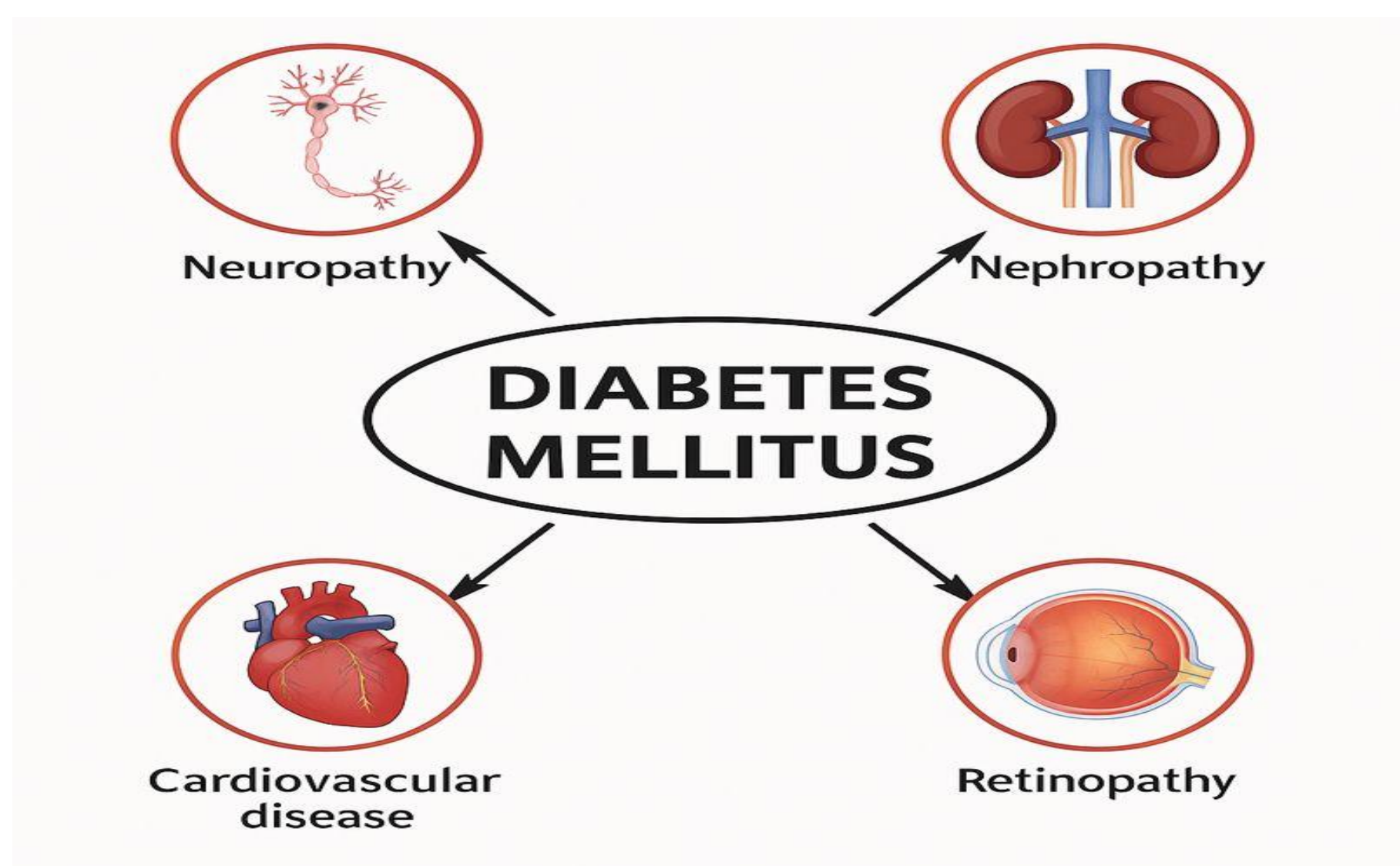


## ABSTRACT

- ML-based multi-class grading of diabetes and identification of predisposing factors.
- BRFSS 2015 “diabetes\_012” dataset: 253,680 adults, 21 variables, strong class imbalance.
- Preprocessing: duplicate removal, outlier filtering ( $|z| > 3$ ), z-score normalization, 70/30 stratified split, SMOTE on training.
- Key predictors (common across 5 selection methods): HighBP, HighChol, BMI, GenHlth, Age.
- Best model: XGBoost (Accuracy 83.11%, Macro-F1 0.42) with good diabetes separation and moderate prediabetes separation.

## INTRODUCTION

Diabetes mellitus is a persistent global health burden and a leading cause of long-term complications affecting multiple organ systems. Early risk assessment and identification of predisposing factors are essential to support prevention and timely intervention, especially in large populations. In this work, we use the CDC’s BRFSS 2015 health survey indicators to develop a machine learning framework for multi-class grading of diabetes status (no diabetes/gestational only, prediabetes, and diabetes) and to identify a compact, interpretable set of key risk-related factors for prediction.



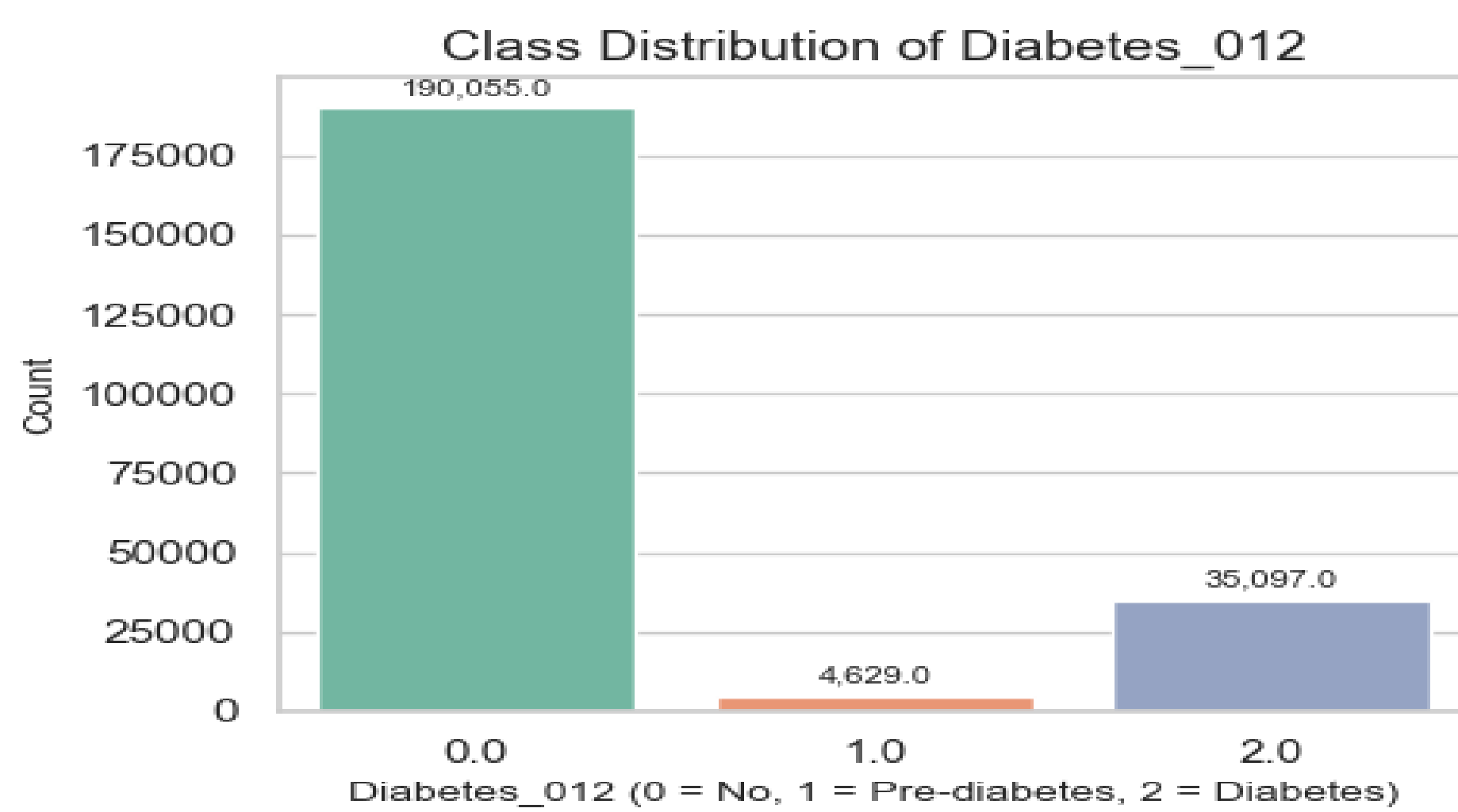
**Figure 1:** Diabetes-related complications motivate the need for early risk assessment and scalable screening approaches.

## OBJECTIVES

- To predict diabetes status using a three-class framework (No Diabetes, Prediabetes, and Diabetes)
- To identify stable predisposing factors associated with diabetes
- To compare multiple machine learning models for multi-class grading

## DATASET

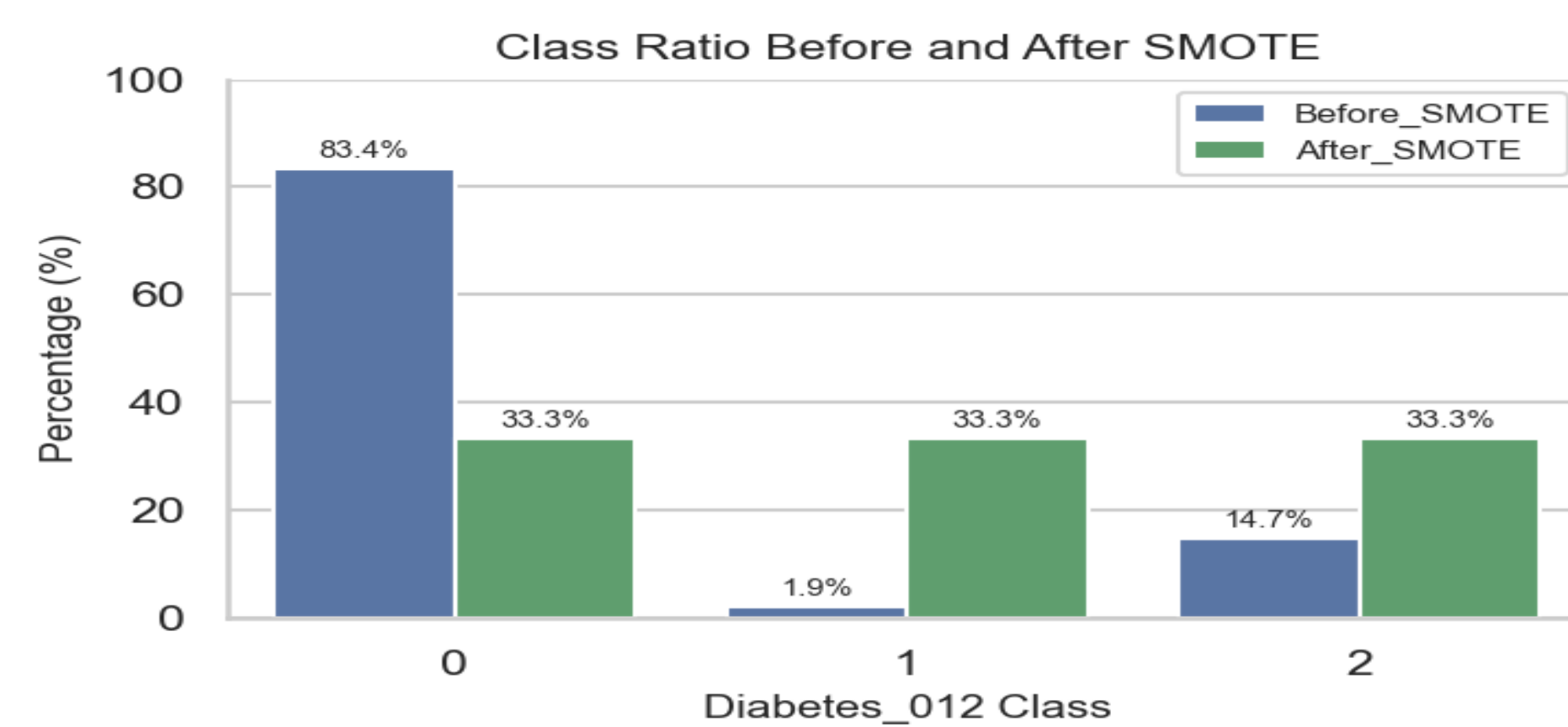
- Source:** Behavioral Risk Factor Surveillance System (BRFSS) 2015
- Target Variable:** Diabetes\_012
  - 0 = No diabetes
  - 1 = Prediabetes
  - 2 = Diabetes
- Features:** Health conditions, lifestyle factors, and demographic indicators



**Figure 2:** Distribution of Diabetes Status

## METHODOLOGY

- Data Preprocessing**
  - Duplicate records removed
  - Outliers removed
  - Numerical features normalized using **Z-score normalization**
- Train-Test Split**
  - 70% training, 30% testing
  - Stratified sampling to preserve class proportions
- Class Balancing**
  - SMOTE applied on the training data only to address class imbalance



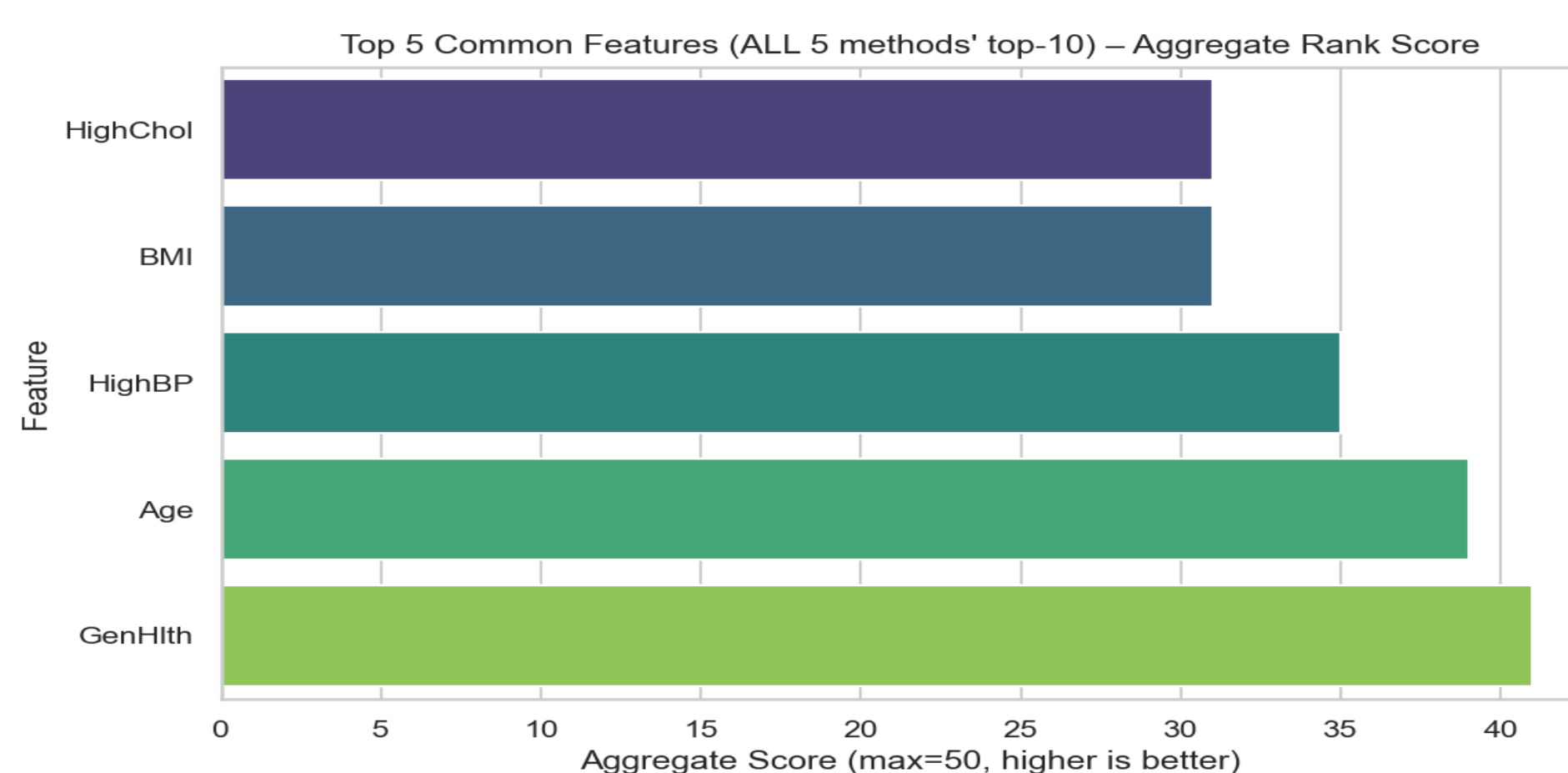
**Figure 3:** Class Ratio Before and After SMOTE:

## FEATURE SELECTION

Feature importance was evaluated using multiple methods:

- Logistic Regression coefficients
- Recursive Feature Elimination (RFE)
- Random Forest importance
- Mutual Information
- Chi-square test

Only stable features selected by multiple methods were used for final modeling.



**Figure 4:** Top 5 Common Predictive Features

## ML MODELS

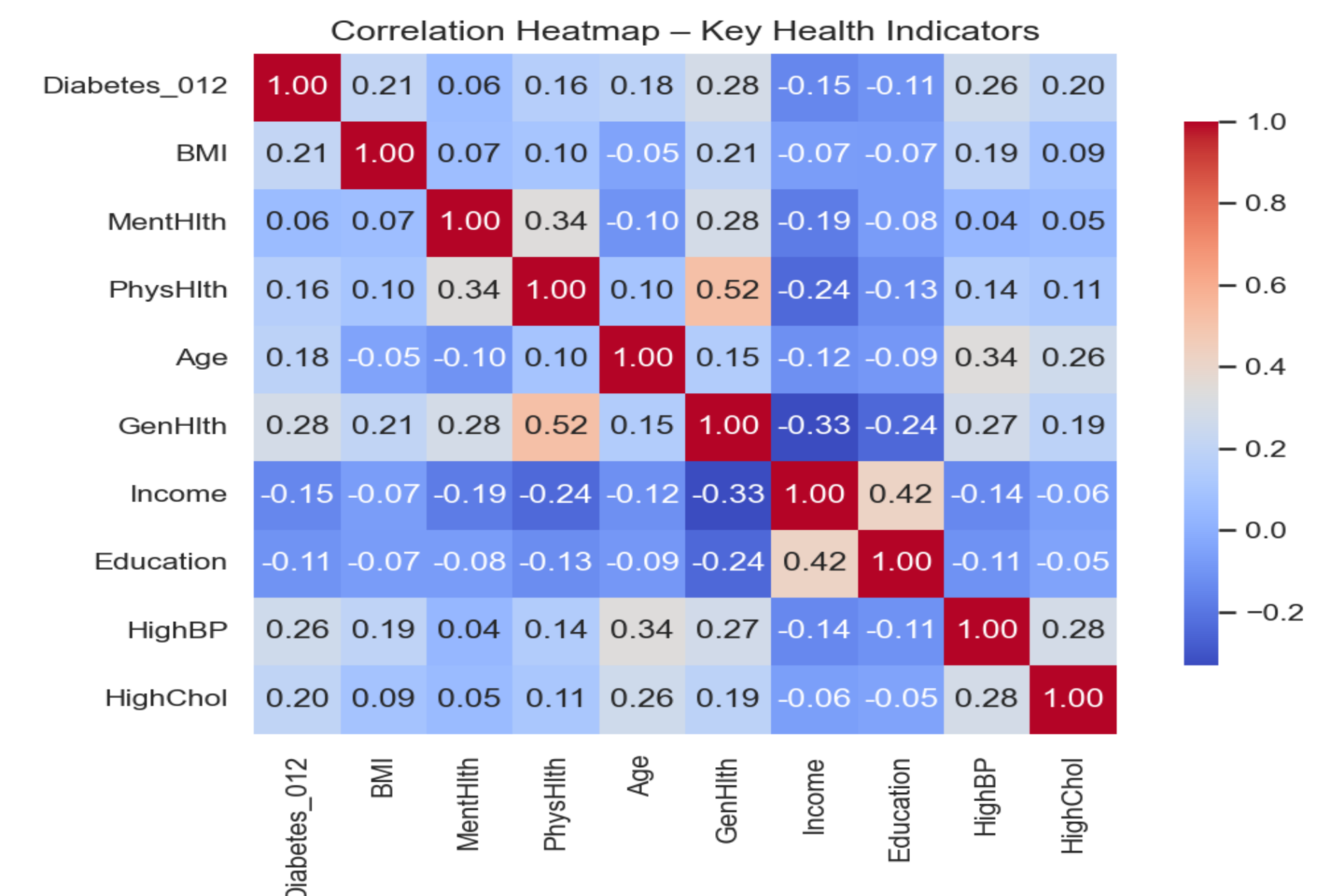
The following models were trained and evaluated:

- Logistic Regression (Multinomial)
- Random Forest
- Support Vector Machine (RBF Kernel)
- XGBoost
- K-Nearest Neighbors (KNN)

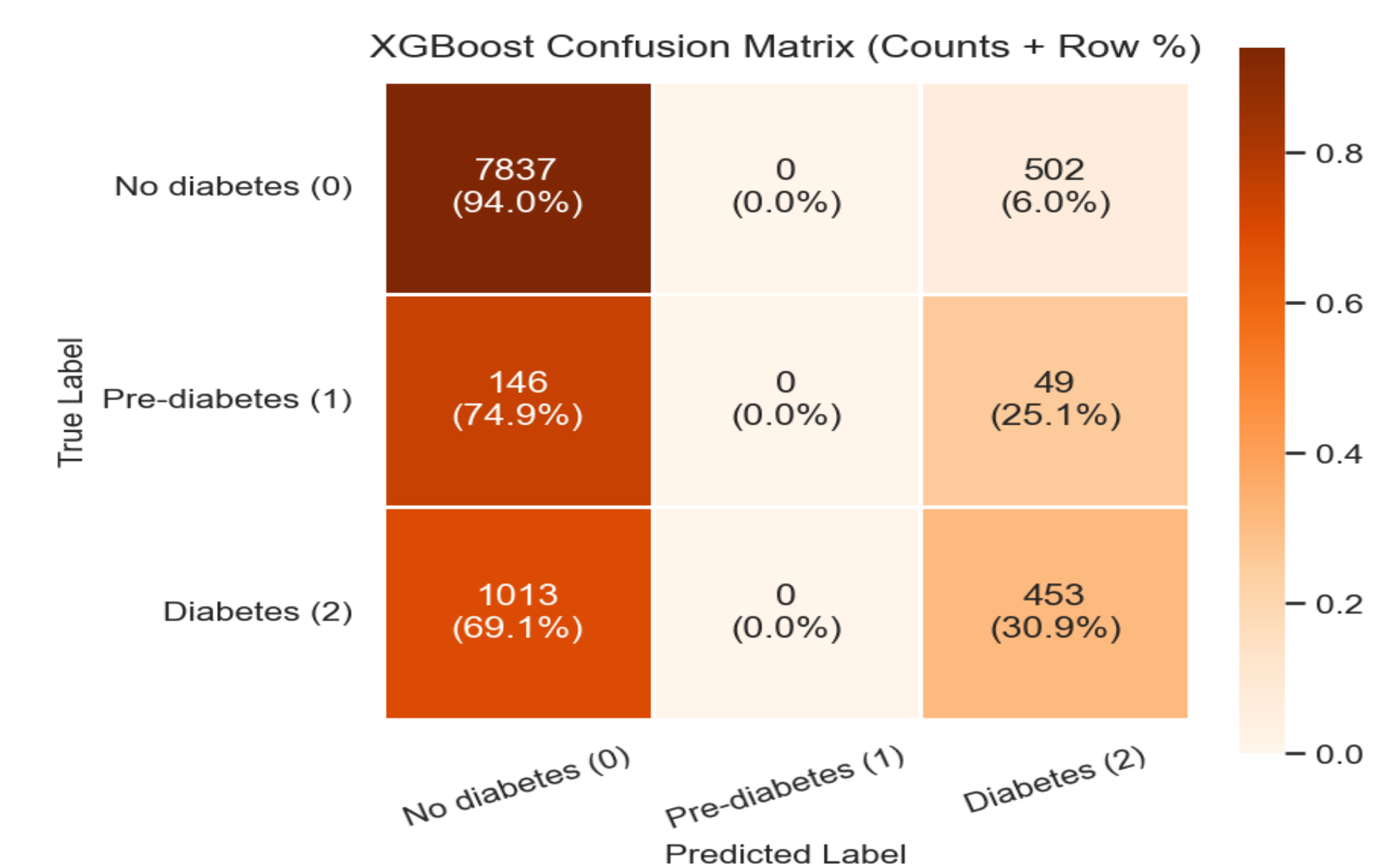
### Evaluation Metrics

- Accuracy
- Macro F1-score
- Confusion Matrix

## RESULTS & DISCUSSION



**Figure 5:** Correlation Heatmap of Key Health Indicators



**Figure 6:** Confusion Matrix for XGBoost Multi-Class Prediction



**Figure 7:** Performance Comparison of ML Models

## CONCLUSION

- Developed a machine learning framework for three-class diabetes grading using BRFSS 2015 data.
- Applied cleaning, normalization, and SMOTE to address data quality and class imbalance.
- Identified stable predisposing factors: HighBP, HighChol, BMI, GenHlth, and Age.
- The best model (by test accuracy & macro-F1) provides a practical baseline for population-level diabetes risk screening.

## FUTURE WORK

- Use real clinical or hospital data
- Apply advanced explainability techniques (SHAP)
- Validate models on local (Bangladesh) datasets
- Explore deep learning models

